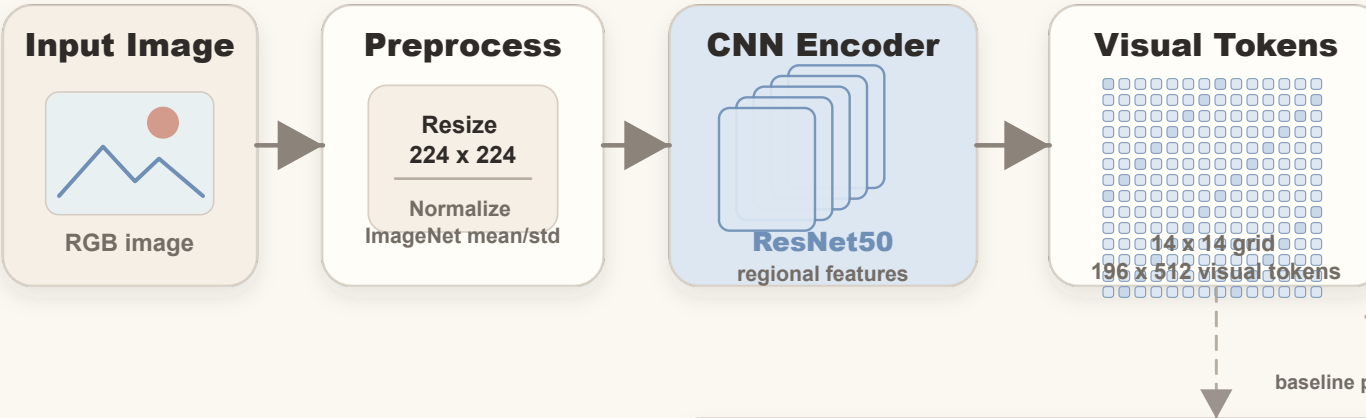


CaptionLens: Semantic Slot Architecture

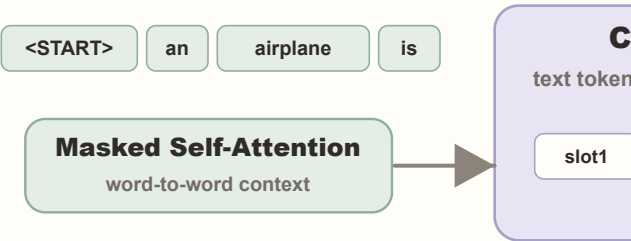
ResNet50 Encoder + Semantic Slot Memory Adapter + Transformer

(a) End-to-end image captioning pipeline



(b) Transformer Decoder with Enhanced Self-Attention

Self-Attention models word history; Cross-Attention aligns visual tokens with text tokens



(c) Empirical gain on MSCOCO 2017 validation

Semantic slots compress visual regions into high-level memory.
Same 20-epoch setting as baseline.

BLEU-4

0.2821 -> 0.2849 (+0.0028)